Dynamics in Mechanics and Biology

1. The Planar Pendulum
   - numerically solving ordinary differential equations
2. Damped Harmonic Motion
   - the ordinary differential equation
   - case analysis
3. Population Dynamics
   - modeling population growth
   - growth with a predator
   - nondimensional terms and steady states
   - catastrophic jumps and hysteresis

MCS 472 Lecture 30
Industrial Math & Computation
Jan Verschelde, 30 March 2026

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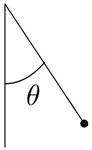
first order equations

Introducing an auxiliary variable $v(t) = \theta'(t)$,

we transform $\theta''(t) + \alpha \sin(\theta(t)) = 0$, $\alpha = g/L$,

into a system of first order differential equations:

$$\theta'(t) = v(t)$$
$$v'(t) = -\alpha \sin(\theta(t))$$



with initial conditions: $\theta(0) = \pi/6$ and $v(0) = 0$.

the package `DifferentialEquations.jl`

```
julia> using DifferentialEquations
```

`DifferentialEquations.jl` is a Julia package which exports a numerical solver for ordinary differential equations.

Christopher Rackauckas and Qing Nie:
Differentialequations.jl – a performant and feature-rich ecosystem for solving differential equations in julia.
Journal of Open Research Software 5(1), 2017.

defining the right hand side function

```
"""
    function pendrhs!(du, u, p, t)

defines the right hand side of the system of first order
order differential equations of a model of a planar pendulum.

The function updates the two components of du,
which respectively define the velocity and the acceleration,
while u[1] contains the value for the angle
and u[2] the value of the velocity.
The p is the array with the parameters
and t is the independent time variable.
"""
function pendrhs!(du, u, p, t)
    angle, velocity = u
    alpha = p
    du[1] = velocity           # instead of u[2]
    du[2] = -alpha*sin(angle) # angle is u[1]
end
```

initial conditions, time span, and parameter

1. The initial conditions are $\theta_0 = \pi/6$, $v_0 = 0$.

```
theta0 = pi/6  # initial angle
v0 = 0 # initial velocity
initconds = [theta0, v0]
```

2. The time span is $[0, 10]$.

```
timespan = (0.0, 10.0)
```

3. The value for the parameter $\alpha = 1$.

```
alpha = 1.0
```

Then we can define the problem:

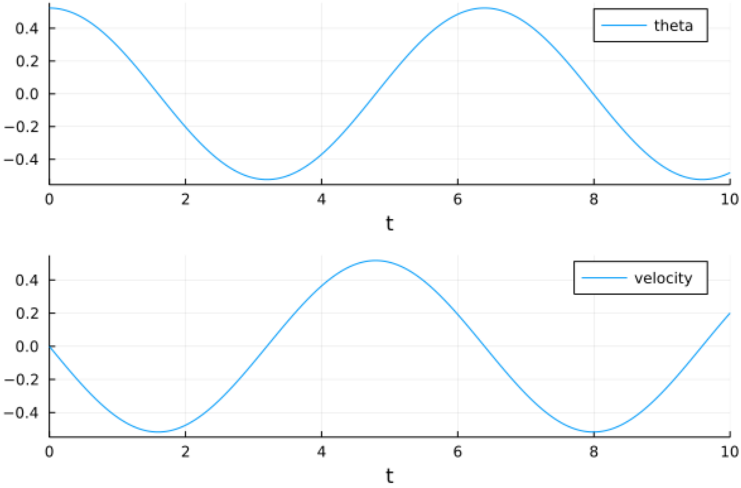
```
ode = ODEProblem(pendrhs!, initconds,
                 timespan, alpha)
```

solving and plotting

```
sol = solve(ode)

plot(sol, xaxis="t",
     label=["theta" "velocity"], layout=(2,1))
```

the plotted solution



the pendulum with damping

The extended model of the pendulum adds damping:

$$\frac{d^2\theta}{dt^2} + A\frac{d\theta}{dt} + \alpha\sin(\theta) = 0, \quad A = 1/10,$$

with initial values $\theta(0) = \pi/6$, $\theta'(0) = \pi/6$.

Exercise 1:

- Define the ODE to solve with `DifferentialEquations.jl`.
- Solve the ODE and plot the trajectories of angles and velocities.

the pendulum with damping and forcing

The extended model of the pendulum adds damping and forcing:

$$\frac{d^2\theta}{dt^2} + A\frac{d\theta}{dt} + \alpha\sin(\theta) = B\sin(t), \quad A = 1/10, \quad B = 10,$$

with initial values $\theta(0) = \pi/6$, $\theta'(0) = \pi/6$.

Exercise 2:

- Define the ODE to solve with `DifferentialEquations.jl`.
- Solve the ODE and plot the trajectories of angles and velocities.

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damped harmonic motion

A model of a suspension mechanism, using the equation for damped harmonic motion, written in engineering notation:

$$\frac{d^2}{d\,t^2} y(t) + 2\zeta\omega \left(\frac{d}{d\,t} y(t) \right) + \omega^2 y(t) = 0.$$

The symbolic solution is

$$y(t) = C_1 e^{\omega t \left(-\zeta - \sqrt{\zeta^2 - 1} \right)} + C_2 e^{\omega t \left(-\zeta + \sqrt{\zeta^2 - 1} \right)}.$$

Observe the discriminant $\zeta^2 - 1 = 0$.

To determine C_1 and C_2, we impose the initial conditions:

$$y(0) = 1 \quad \text{and} \quad y'(0) = 0.$$

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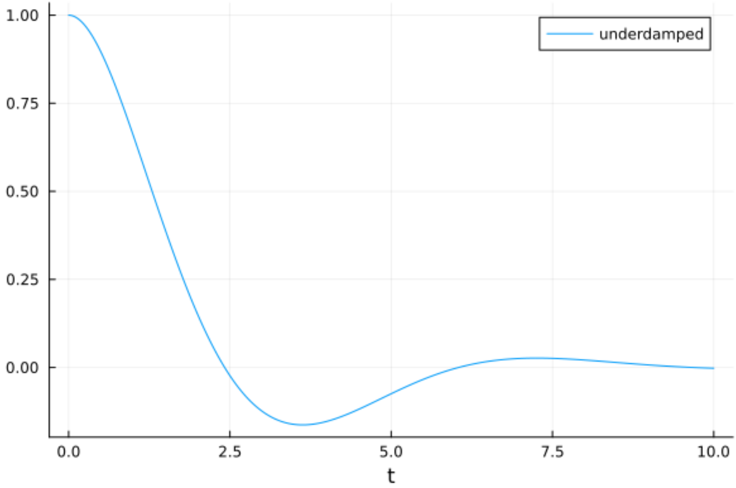
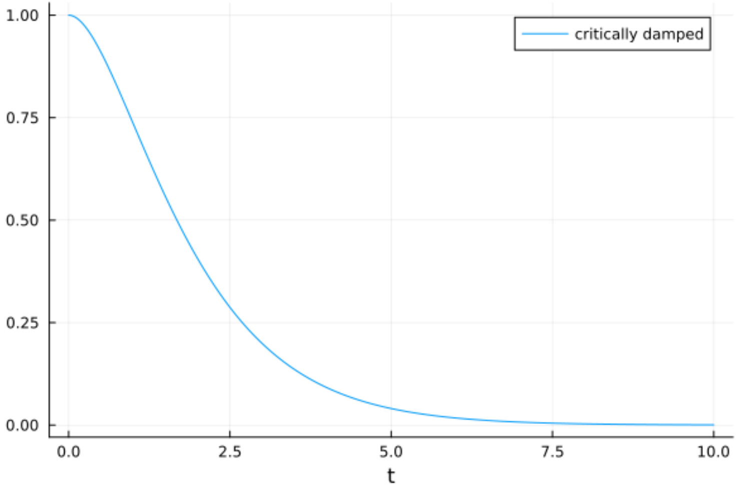
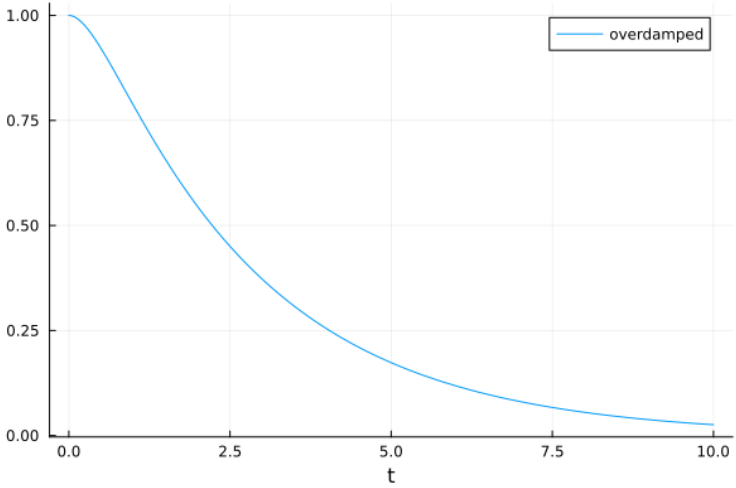
underdamped, $\zeta = 1/2$, $\omega = 1$



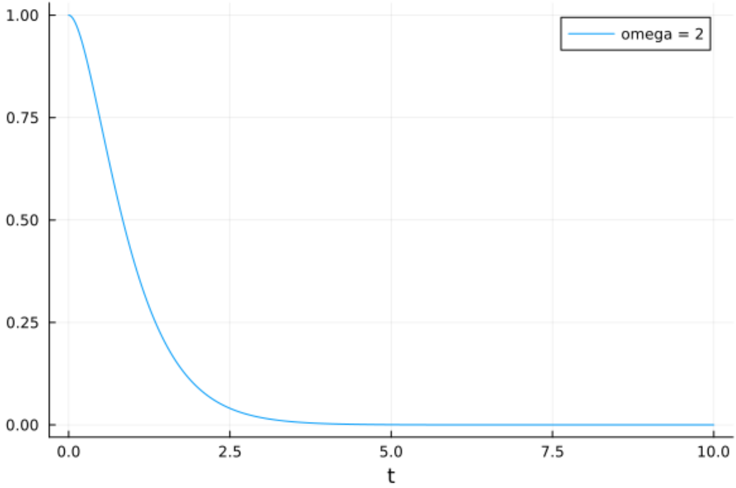
critically damped, $\zeta = 1$, $\omega = 1$



overdamped, $\zeta = 3/2$, $\omega = 1$



damped faster, $\zeta = 1$, $\omega = 2$



how long to decay?

Exercise 3: Consider

$$\frac{d^2 y}{d\,t^2} + \frac{1}{10}\frac{d\,y}{d\,t} + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

- Compute the solution trajectory for this problem.
- How long before $y(t) \leq 0.01$?

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modeling population growth

A differential equation to model the growth of a population was proposed by Verhulst in the 19th century.

We use the following notation:

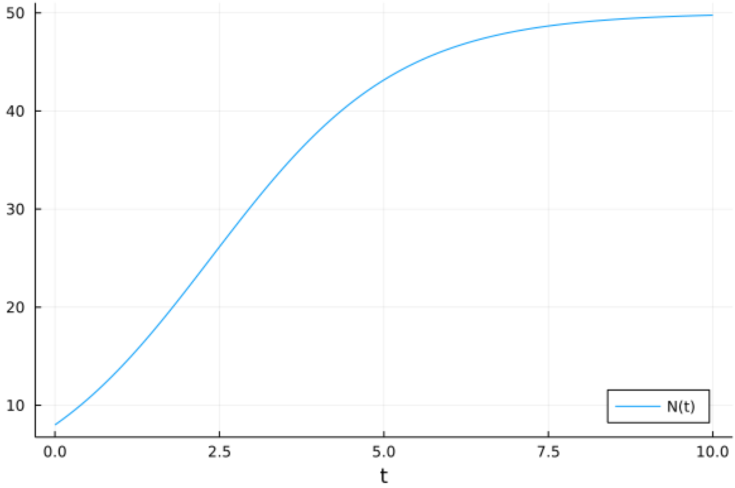
- Let $N(t)$ be the number of the population in function of time t.
- The carrying capacity is K.
- The birth rate is r.

$$\frac{d}{dt} N(t) = r \left(1 - \frac{N(t)}{K} \right) N(t).$$

The rate of increase in population

- is proportional to the birth rate times the population number,
- multiplied with a term to take the carrying capacity into account.

logistic growth $K = 50$, $N_0 = 8$, $r = 0.7$



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growth with a predator

We consider the population dynamics of the spruce budworm.
Birds eat worms.

From the logistic growth model we subtract a predator term.
The predator term is a function of the population number: $P(N(t))$.

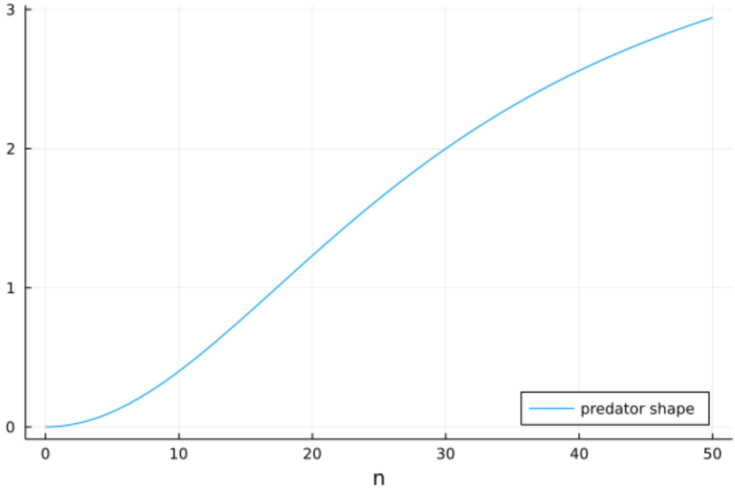
$$\frac{d}{dt} N(t) = r \left(1 - \frac{N(t)}{K} \right) N(t) - P(N(t)).$$

The population of the worms should reach a critical value
before the birds start noticing the worms.

Once this happens, the birds eat worms in greater numbers.

Once the population falls below another critical value,
the predation disappears.

the predator term $Bn^2/(A^2 + n^2)$, $A = 30$, $B = 4$



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nondimensional terms

Our model

$$\frac{d}{dt}N(t) = r\left(1 - \frac{N(t)}{K}\right)N(t) - \frac{B(N(t))^2}{A^2 + (N(t))^2}$$

has too many parameters.

Applying the substitutions

$$N(t) = Au(t), \quad r = \frac{Br}{A}, \quad K = qA$$

leads to

$$\frac{d}{dt}Au(t) = Br\left(1 - \frac{u(t)}{q}\right)u(t) - \frac{A^2B(u(t))^2}{A^2 + A^2(u(t))^2}.$$

Observe that the A^2 cancels in the predator term.

two more substitutions

With $u(t) = u(tB/A)$ we can remove B

$$\frac{d}{dt} A u(tB/A) = Br \left(1 - \frac{u(tB/A)}{q} \right) u(t) - \frac{A^2 B (u(tB/A))^2}{A^2 + A^2 (u(tB/A))^2}$$

because B appears at the left and the right of the equation.

Then we replace Bt/A by t:

$$\frac{d}{dt} u(t) = r \left(1 - \frac{u(t)}{q} \right) u(t) - \frac{(u(t))^2}{1 + (u(t))^2},$$

with

$$r = \frac{B}{A} r \quad \text{and} \quad q = \frac{K}{A}.$$

steady states

At the steady states of

$$\frac{d}{dt}u(t) = r\left(1 - \frac{u(t)}{q}\right)u(t) - \frac{(u(t))^2}{1 + (u(t))^2}$$

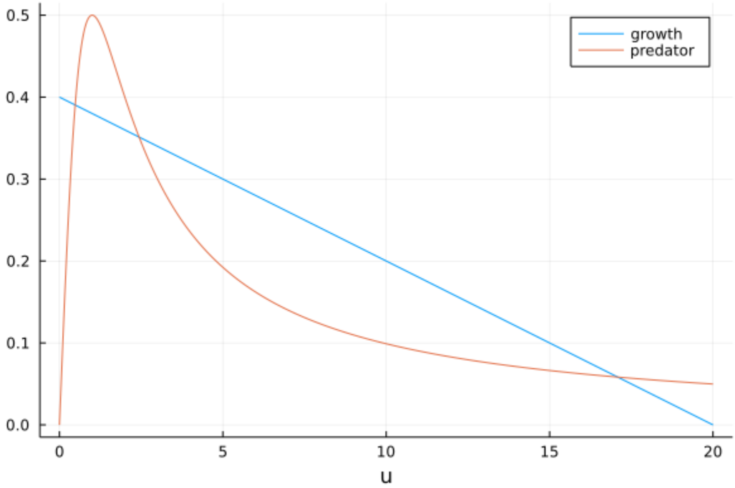
the right hand side equals zero.

Replace $u(t)$ by x and we find

$$r\left(1 - \frac{x}{q}\right)x - \frac{x^2}{1 + x^2} = 0.$$

Observe that $x = 0$ is a trivial factor.

steady states for $r = 0.4$ and $q = 20$



numerical roots

Applying `solve()`, for $r = 0.4$ and $q = 20$ gives

```
3-element Vector{Sym}:
 0.480535939911864 - 0.e-22*I,
  2.4363330031605 5 + 0.e-19*I,
  17.0831310569276 - 0.e-20*I
```

computing a special model

Symbolic solving of

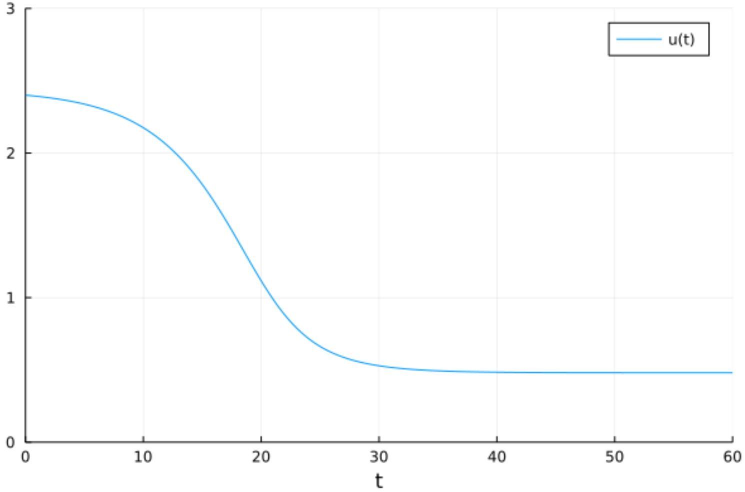
$$\frac{d}{dt}u(t) = 0.4\left(1 - \frac{u(t)}{20}\right)u(t) - \frac{(u(t))^2}{1 + (u(t))^2}$$

is not possible. We use `DifferentialEquations.jl`.

```
"""
    function wormrhs!(du, u, p, t)

defines the right hand side for a special worm model.
"""
function wormrhs!(du, u, p, t)
    growth = 0.4*(1 - u[1]/20)*u[1]
    predator = - (u[1]^2)/(u[1]^2 + 1)
    du[1] = growth + predator
end
```

a special model



We started at 2.4, very close to a steady state by ended up at 0.4.

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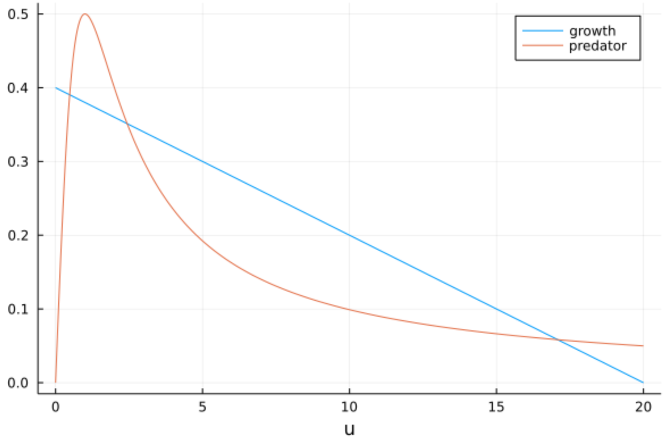
catastrophic jumps



As r increases from 0.4 to 0.5 while $q = 20$ remains fixed, the two steady states become closer to each other and then disappear.
As r increases, we get a catastrophic outbreak of the worm population.

catastrophic jumps

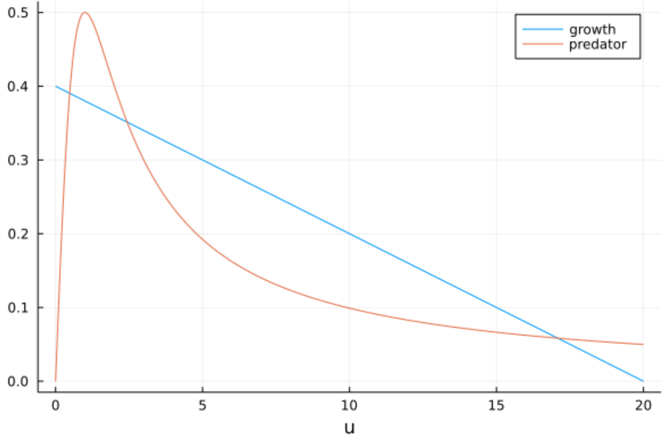
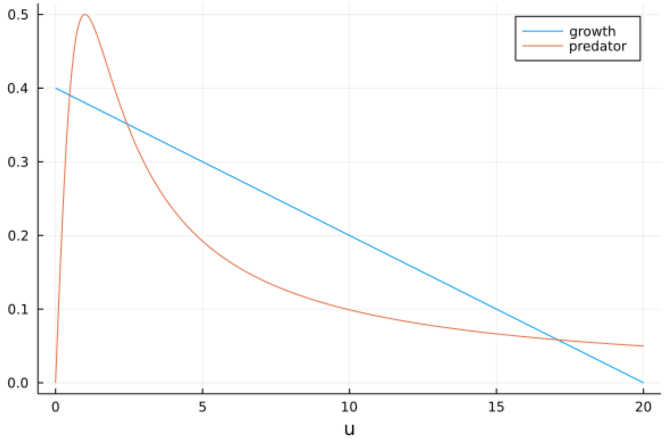


Image now that we would start at $r = 0.6$, at a large steady state. The return to a smaller value of the population will only be possible when the large steady state, jointly with the unstable middle steady state disappears as the predator line sinks below the growth line.

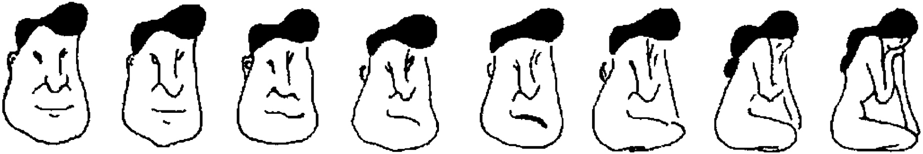
catastrophic jumps and hysteresis



The catastrophic jumps happen at different values of the parameter r and are depending on the direction of the parameter change.
This is called *hysteresis*.

a catastrophe in visual perception

Look at the pictures below from left to right



and then from right to left.

Where the sudden jump in perception happens depends on the order, whether we look left-to-right or right-to-left.

another specific model

Exercise 4: Consider $u(t)$ as the solution of

$$\frac{d\,u}{d\,t} = r\,u\left(1 - \frac{u}{q}\right) - \frac{u^2}{1+u^2},$$

where r and q are parameters with positive values.

1. Show that the regions for which there is 1 positive steady state, or 3 positive steady states, are described by

$$r = \frac{2a^3}{(1+a^2)^2} \quad \text{and} \quad q = \frac{2a^3}{a^2-1},$$

   for some parameter a.

2. Solve the problem for $a = \sqrt{3}$ and make a plot.
3. Interpret the solution, for when $a \rightarrow \infty$?
4. Interpret the solution, for when $a \rightarrow 1$?

population model with age distribution

Proposal for a Topic of a Project:

One deficiency of population models is that they do not take the age structure into account.

Read section 1.7 of *Mathematical Biology I. An Introduction* by James D. Murray, Springer-Verlag 2002.

Consider the following.

- Work through section 1.7 computationally and build a numerical model to reproduce the plots in that section.
- What is the effect of an aging population on its growth?

summary and bibliography

With symbolic and numeric computation, we solved ordinary differential equations, which in mechanics and biology.

We are in the middle of Chapter 9 of our text book.

- Charles R. MacCluer: *Industrial Mathematics. Modeling in Industry, Science, and Government*. Prentice Hall, 2000.

The population dynamics is based on

- James D. Murray: *Mathematical Biology I. An Introduction*. Third Edition, Springer-Verlag 2002.